Ganesh Gupta

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Experience

Oracle | *Software Engineer, AI Platform* Jun 2025 – Present

- Built observability and autoscaling for **vLLM** inference serving with **Prometheus/Grafana**, and trained and fine-tuned **PyTorch** models on multi-GPU clusters.
- Developed and deployed **deep learning models** in **PyTorch** and **TensorFlow** with automated training workflows and **MLOps CI/CD** on **AWS SageMaker**.
- Provisioned and operated multi-cloud **GPU clusters** (AWS, GCP, CoreWeave) with **NVIDIA/CUDA** and **InfiniBand**, running distributed training on **Slurm** and **Kubernetes**.
- Led a production **multi-agent system** on **OpenAI Agents SDK**, **LangGraph**, and **MCP** tool use that autonomously processes **financial workflows**, handling **millions of daily transactions** at **99%+ uptime**.

QureAI | *ML Infrastructure Engineer* Mar 2026 – Present

- Moved LLM serving to an **async, autoscaled** architecture with **TensorRT-LLM**, cutting **p99 latency from 420ms to 140ms** across **50K+ daily requests**.
- Worked on **vLLM-class** serving internals including distributed **KV cache** and **speculative decoding** to raise inference throughput.
- Built **PySpark** jobs and **Airflow** DAGs for large-scale feature and dataset preparation feeding training and **evals**.

Nomura Research Institute Financial Technologies | *Software Engineer* Sep 2019 – Aug 2023

- Built **Java**, Python, and **C++** services, standardizing retry, timeout, and failure-propagation contracts to cut production escalations **60%**.
- Architected and operated **25+ production Python and Java microservices** for **1M+ users** in **financial services** at **99.9% uptime** on **AWS**.
- Cut nightly batch aggregation from **8 minutes to 90 seconds** by optimizing **SQL and PL/SQL** on a financial RDBMS with indexed views and partitioning.
- Led **architecture and code reviews** and mentored engineers on system design, reliability, and secure API practices.

Projects

Distributed KV Store `github.com/ganeshhgupta/distributed-kv-store`

- **Raft**-consensus key-value store with **linearizable reads** and **log compaction**, plus a companion VPC control plane simulator.

Watch-your-LLM `github.com/ganeshhgupta/watch-your-llm`

- **LLM evaluation** platform with rubric-based scoring, **hallucination detection**, regression alerting, a **FastAPI** backend, and a **React/TypeScript** dashboard.

RootCause AI `github.com/ganeshhgupta/rootcause-ai`

- **LangGraph** multi-agent system that routes production alerts to specialized sub-agents via **tool-use** chains and writes structured root-cause reports, cutting **MTTR 65%**.

Education

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0* Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science* Aug 2016 – Jul 2020

Skills

Languages: C++, Python, Java, Go, SQL

AI Runtime & Inference: LLM Inference, vLLM, TensorRT-LLM, KV Cache, Speculative Decoding, Quantization, Model Compression

ML: PyTorch, TensorFlow, DNNs, LLMs, Performance and Accuracy Validation

Tools: Docker, Kubernetes, CUDA, Git, CI/CD, Prometheus/Grafana

Certifications: AWS Certified Developer Associate, GCP Professional MLOps Engineer